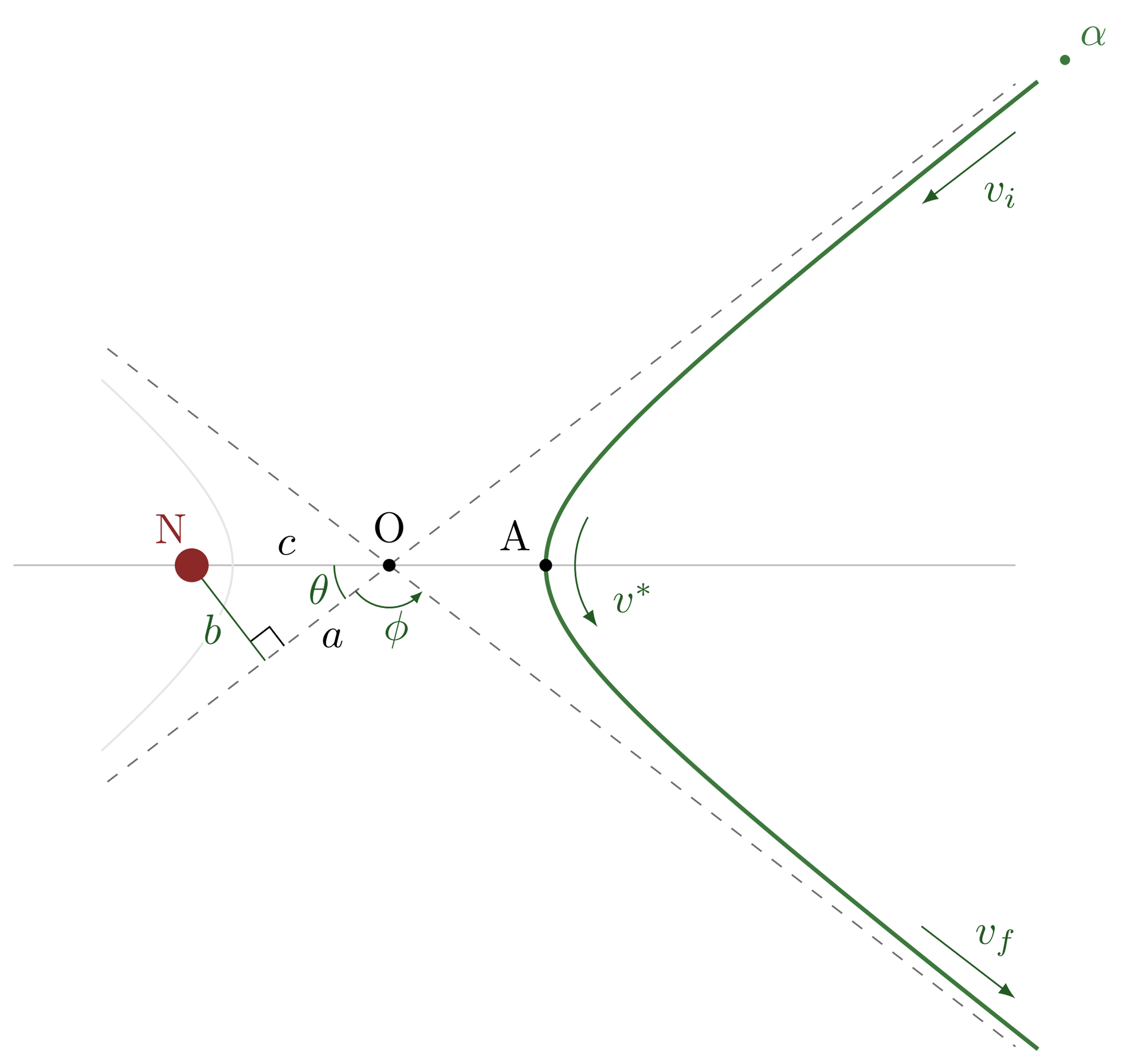
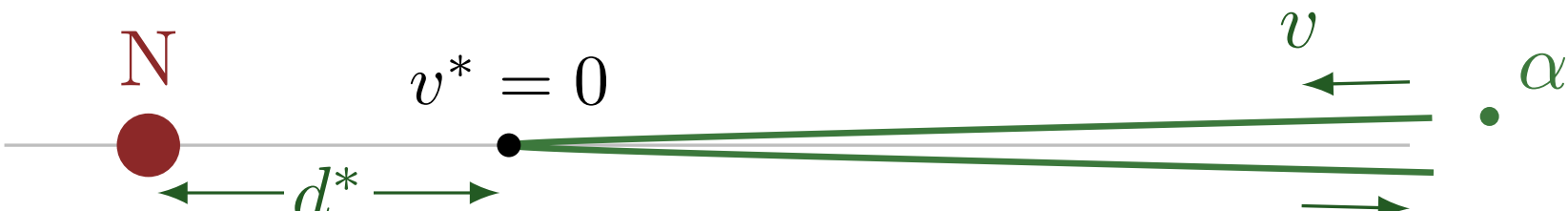
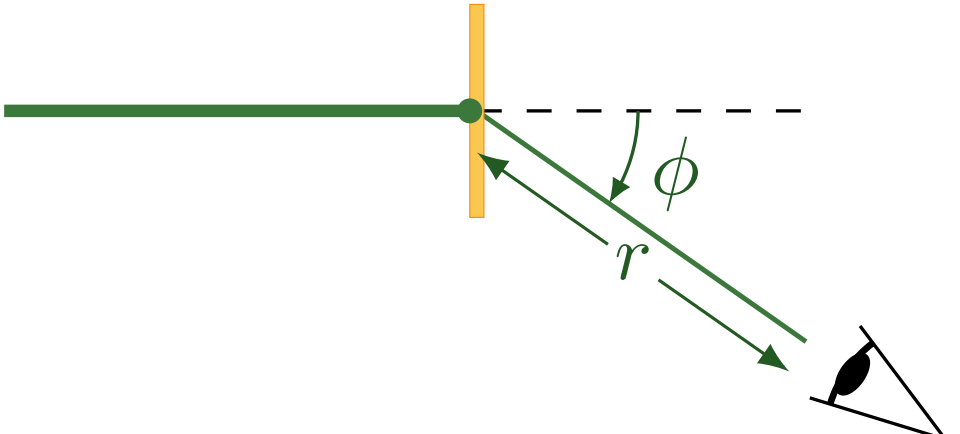
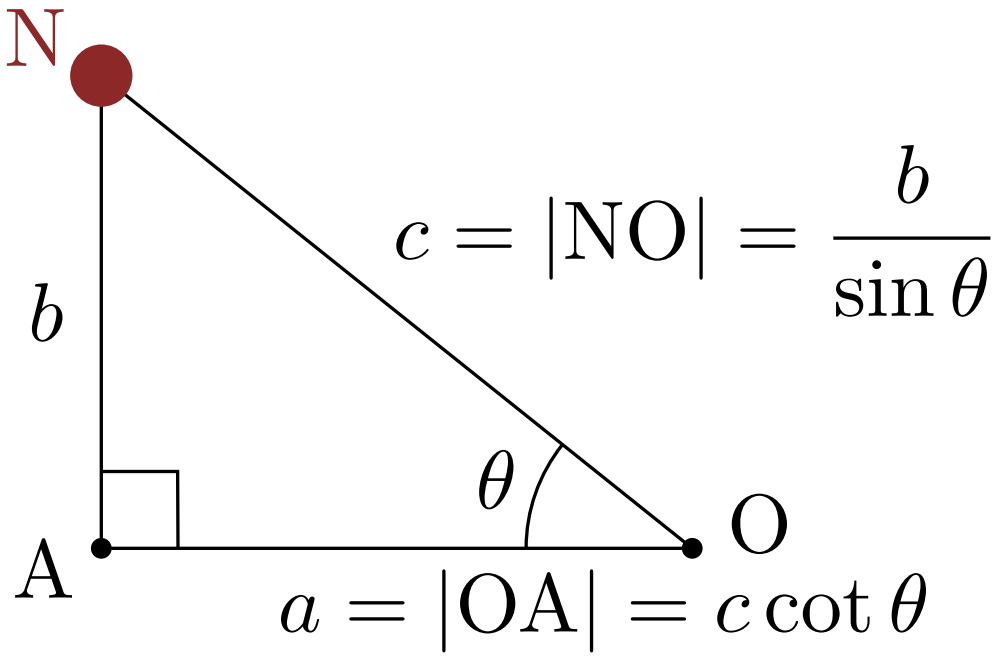
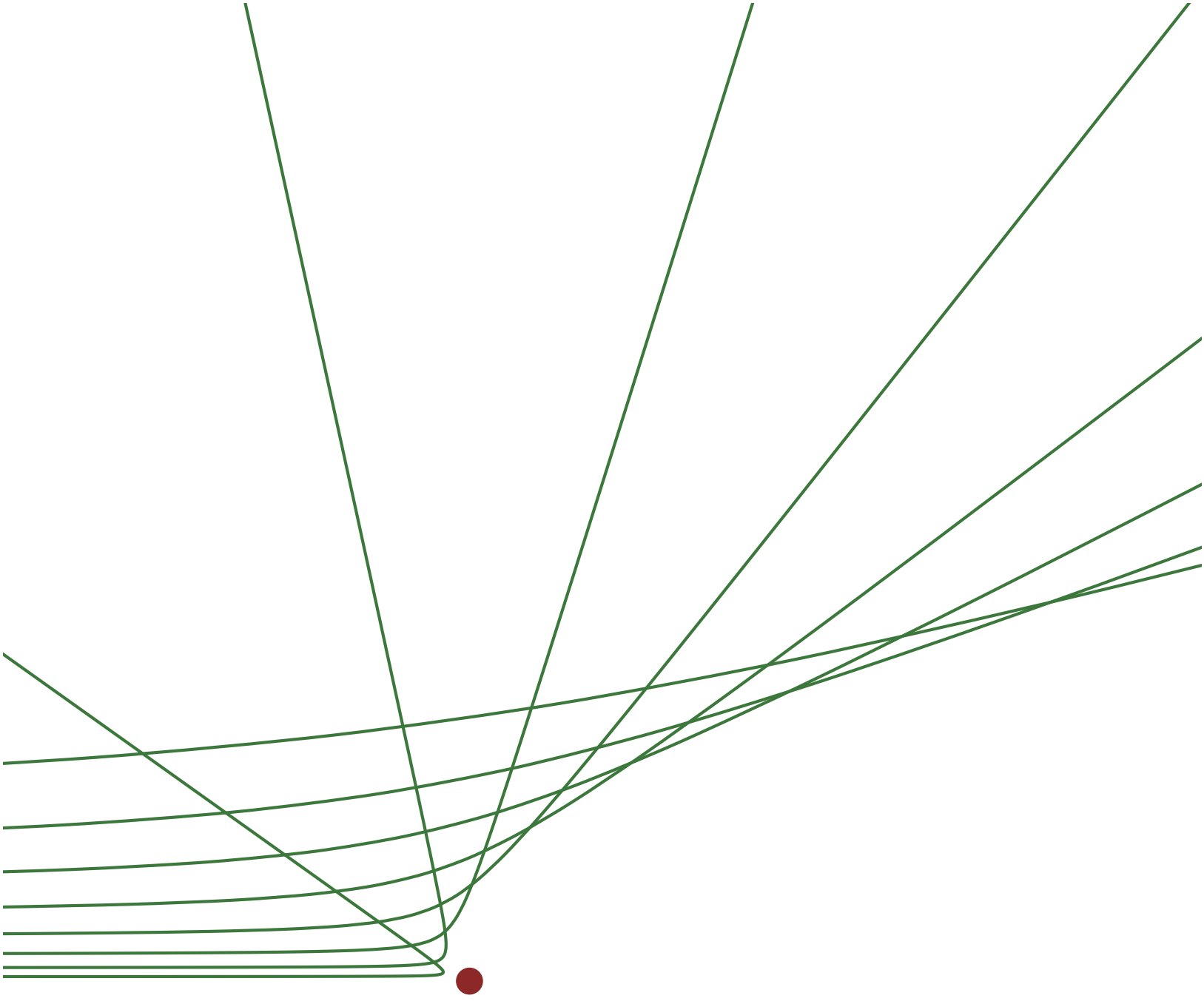
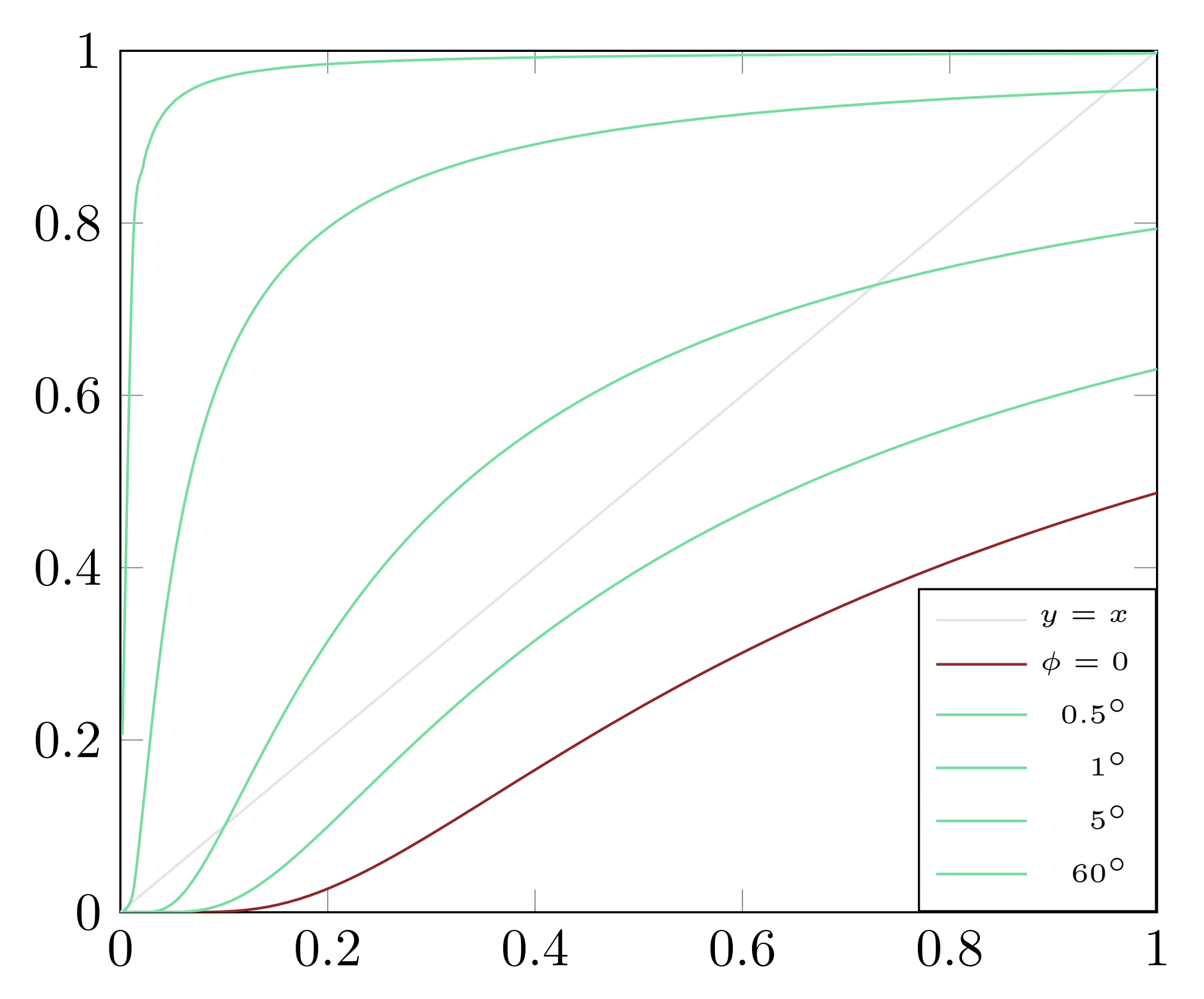

α
v_i
N
c
O
A
θ
ϕ
v^*
b
a
v_f


N
$v^* = 0$
v
α
d^*


ϕ
r


N
$c = |\mathrm{NO}| = \dfrac{b}{\sin\theta}$
b
θ
O
A
$a = |\mathrm{OA}| = c\cot\theta$




1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1
$y = x$
$\phi = 0$
0.5°
1°
5°
60°